

Typst Smoke Test Worksheet

AP Calculus AB
Mr. Test
Academic Worksheet

Name:

Date: 2026-03-12

Class:

Score:

Practice

完成以下题目并保留必要步骤。

1. 3 分

What is the derivative of $f(x) = x^2 + 3x$?

A $2x + 3$

B $x^2 + 3$

C $2x$

D $3x + 2$

2. 3 分

Which of the following statements about limits is true?

- A** If $\lim_{x \rightarrow a} f(x)$ exists, then $f(a)$ must be defined
- B** If $f(a)$ is defined, then $\lim_{x \rightarrow a} f(x)$ must exist and equal $f(a)$
- C** The limit of a function at a point can exist even if the function is not defined at that point

3. 5 分

Find the area under the curve $y = x^2$ from $x = 0$ to $x = 3$. Show your work using the Fundamental Theorem of Calculus.

Answer Key

仅供教师参考

1. A

解析

$f'(x) = 2x + 3$, 由幂法则和常数求导可得。

2. C

解析

极限可以在函数于该点未定义时存在, 例如 $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ 。

3. 9

解析

$$\int_0^3 x^2 dx = \left[\frac{x^3}{3} \right]_0^3 = \frac{27}{3} - 0 = 9$$